Blockchain-Empowered Decentralized Horizontal Federated Learning for 5G-Enabled UAVs

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***Abstract*—Motivated by Industry 4.0, 5G-enabled unmanned aerial vehicles (UAVs; also known as drones) are widely applied in various industries. However, the open nature of 5G networks threatens the safe sharing of data. In particular, privacy leakage can lead to serious losses for users. As a new machine learning paradigm, federated learning (FL) avoids privacy leakage by allowing data models to be shared instead of raw data. Unfortunately, the traditional FL framework is strongly dependent on a centralized aggregation server, which will cause the system to crash if the server is compromised. Unauthorized participants may launch poisoning attacks, thereby reducing the usability of models. In addition, communication barriers hinder collaboration among a large number of cross-domain devices for learning. To address the abovementioned issues, a blockchain-empowered decentralized horizontal FL framework is proposed. The authentication of cross-domain UAVs is accomplished through multisignature smart contracts. Global model updates are computed by using these smart contracts instead of a centralized server. Extensive experimental results show that the proposed scheme achieves high efficiency of cross-domain authentication and good accuracy.**

***Index Terms*—5G-enabled unmanned aerial vehicles (UAVs), cross-domain authentication, federated learning (FL), privacy preservation, smart contract.**

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I. Introduction

THE industrial Internet of Things (IIoT) paradigm is the core driving force of Industry 4.0, which is fundamentally transforming many industries, such as transportation [1], artificial intelligence (AI) [2], and healthcare [3], [4]. IIoT technology uses machine learning (ML) to process the large amount of data generated by connected devices. However, while using data to train ML models is an effective way to take full advantage of the value of those data, it also raises concerns about privacy leakage [5]. In the IIoT context, AI-based 5G-enabled unmanned aerial vehicles (UAVs) are expected to be widely used in transportation, traffic control, social governance, smart healthcare, and other fields in the future [6]. Accordingly, security and privacy protection have become two major challenges for UAV applications [7]. The collaborative use of data has a wide range of applications in distributed scenarios, especially for highly autonomous devices. Traditional ML algorithms utilize data from data centers. However, data owners are often reluctant to share their sensitive data, thus hindering the ML process. Hence, a distributed ML technique called federated learning (FL) [8] has been proposed to avoid data leakage and guarantee the performance of ML. In FL, each data owner obtains a set of local learning parameters, and all data owners then share their local updates with a centralized authority to maintain a global model. The confidentiality of the training data is guaranteed because the data are stored locally; however, the centralized aggregator may suffer from attacks [9]. The research reported in [10] shows that gradient updates may disclose information about a participant's training data and that, consequently, attackers may be able to recover data from the uploaded gradients. In view of this, privacy preservation has aroused extensive concern from both academia and industry [11]. Privacy-preserving techniques, such as secure multiparty computation, homomorphic encryption, and differential privacy, have been successively applied in FL to defend against privacy attacks on models. However, it is difficult to achieve authentication when implementing these techniques for UAVs due to their scattered geographical locations, making the corresponding FL models vulnerable to malicious attacks [12]. In addition, the more participants join in FL, the better the global model will be. While the number of devices in one domain is limited, a better model will be obtained if multiple domains collaboratively participate in training. However, the

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authentication mechanism may vary from one domain to another, making collaborative learning across domains difficult to realize. To address the abovementioned problems, a blockchain-empowered decentralized cross-domain FL (BE-DHFL) framework for 5G-enabled UAVs is proposed in this article. Under this model, only updates uploaded by authenticated UAVs will be added to the model pool for aggregation. The main contributions of this article are as follows.

1) The privacy of data is protected by applying FL to share data models rather than raw data.
2) A distributed cross-domain authentication method is proposed to realize the authentication of cross-domain participants in FL.
3) We propose a decentralized FL framework that utilizes blockchain technology for model aggregation to avoid the single point of failure represented by a centralized aggregator.
4) Security and experimental analyses are conducted to prove the effectiveness of the proposed scheme.

The rest of this article is organized as follows. Section II gives an overview of related studies. Section III presents the system model. The BE-DHFL framework is described in detail in Section IV. Section V analyzes the security and experimental results of the proposed scheme. Section VI concludes this article.

II. Related Work

A 5G network is an open environment, presenting great security challenges for 5G applications from the perspective of privacy preservation [13], [14]. As a new fusion technique, FL enables data owners to collaboratively train a global model without sharing their raw data [15]; however, the centralized server used in traditional FL is semihonest. Aono *et al.* [16] suggested preventing aggregator attacks by using homomorphic encryption. However, the security of the system can be compromised by malicious participants constructing a generative adversarial network (GAN). Weng *et al.* [17] designed a system called DeepChain for collaborative learning, but these authors did not propose a privacy preservation method. Jiang *et al.* [18] presented a collaborative training method to protect feature privacy, but they did not use FL to protect the privacy of the training data. As a mature distributed ledger technology, blockchains are widely exploited in FL to enhance system security and model performance. Awan *et al.* [19] proposed a blockchain-based privacy-preserving framework that employs the tamper-evident nature of a blockchain to guarantee the model updates. Zhao *et al.* [20] presented an FL approach for Internet of Things (IoT) devices and applied blockchain technology to audit the updates, thus preventing the participation of malicious users. Pokhrel *et al.* [21] developed an IoT-oriented FL framework to improve privacy protection and communication efficiency in the Internet of Vehicles. Lu *et al.* [15] extended this pattern to the IIoT; in their scheme, data sharing was achieved by using the blockchain and FL approaches.

In all of the aforementioned FL schemes, a centralized server is required to aggregate the local updates. Unfortunately, however, this configuration is highly susceptible to server failures.

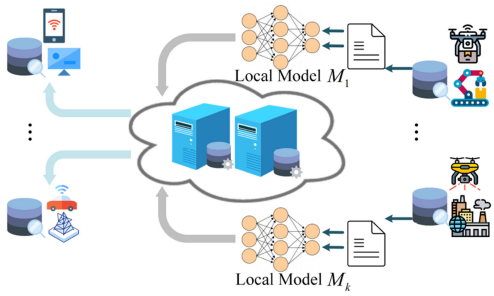


Fig. 1. Example of centralized FL in the IIoT paradigm.

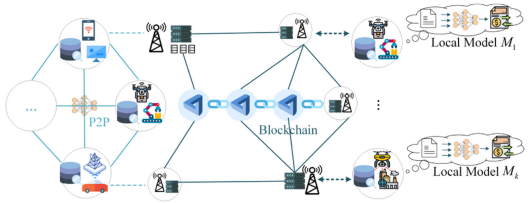


Fig. 2. Example of decentralized FL in the IIoT paradigm.

Qu *et al.* [22] proposed an FL solution using a specially designed blockchain with privacy protocols to replace the centralized server; however, their solution does not include a method of authenticating the legitimacy of the participants. Ma *et al.* [23] recommended the utilization of smart contracts for model aggregation. However, they also did not authenticate the participants. To solve the authentication problem for resource-constrained IoT devices, Shahidinejad *et al.* [24] proposed a safe and efficient lightweight authentication protocol. Shen *et al.* [25] combined blockchain and identity-based signature technology to achieve authentication of cross-domain IIoT devices. Guo *et al.* [26] designed a master–slave blockchain architecture to achieve authentication across different domains in the IoT context. In this work, we optimize the authentication methods proposed in [25] and [26] to make the authentication process more flexible and apply the optimized process in our decentralized FL system.

III. System Model

The variants of the FL framework include horizontal FL, vertical FL, and federated transfer learning. In this article, we consider a scenario in which cross-domain drones participate in FL. The datasets collected by drones tend to have similar data characteristics; therefore, we use horizontal FL. An example of centralized FL in the IIoT paradigm is shown in Fig. 1. Our goal is to establish a secure and effective mechanism for decentralized FL among cross-domain IIoT devices. Suppose that the number of participants is sum and that each participant T_i owns a dataset D_i. Each participant trains a model locally on D_i to obtain a local update m_i for the joint training of a global model $\mathcal{M}$ in collaboration with the other participants. This decentralized FL framework in the IIoT paradigm is illustrated in Fig. 2. In this framework, the verification of the local updates and the generation of the global model are accomplished by using a blockchain or peer-to-peer (P2P) network.

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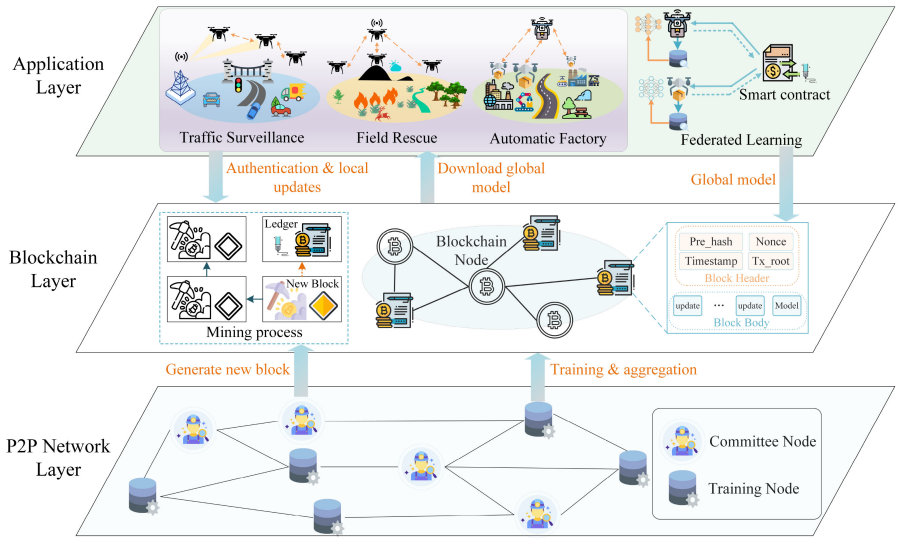


Fig. 3. Proposed BE-DHFL framework.

A. System Architecture

As shown in Fig. 3, the framework of the proposed scheme consists of three layers: an application layer, a blockchain layer, and a P2P network layer. The set of drones is $D = \{d_1, d_2, \ldots, d_n\}$, the set of training nodes is $N_t = \{n_{t,1}, n_{t,2}, \ldots, n_{t,k}\}$, and the set of committee nodes is $N_c = \{n_{c,1}, n_{c,2}, \ldots, n_{c,m}\}$. We exploit a consortium blockchain, which is maintained by the domains involved in FL. In this article, each domain deploys a supernode on certain entities (e.g., base stations or roadside units) with high computing and storage capabilities to maintain the blockchain network. The drones are securely connected through cryptographic records in the blockchain, the participants are authenticated through authentication contracts, and the global model is generated through updating contracts.

B. Application Layer

The application layer includes a key generation center, a large number of IIoT devices, and smart contracts.

1) Key Generation Center (KGC): In each FL domain, there is a unique KGC that is primarily responsible for the identity management of drones. It also acts as the administrator of the blockchain in maintaining smart contracts. Specifically, KGC^N completes both local and global registrations for drones in domain N and dynamically updates the registration information in the blockchain. Local registration is used for resource management within the domain, and global registration is used to achieve cross-domain authentication.

2) Terminals: The terminals include drones and intelligent IIoT devices with sensing, storage, and computing capabilities. They are responsible for completing one or more specific tasks. These terminals need to submit registration information to their KGC, which then completes registration for them in the corresponding private blockchain and the consortium blockchain. The drones participating in each round of learning need to download the models for training from their associated committee nodes.

3) Smart Contracts: In this article, two types of smart contracts are applied: authentication contracts and updating contracts. An authentication contract is used to implement cross-domain authentication, and an updating contract is used for model aggregation.

C. Blockchain Layer

A blockchain network is a distributed network composed of authorized and unauthorized nodes that can securely interact with each other for data sharing. Based on the distributed nature and encryption characteristics of blockchain technology, such a system can establish a foundation of mutual trust among different entities. A new block is created after that block is filled or completed. All blocks are linked in chronological order, and each block contains a hash of the previous block. We exploit a consortium blockchain as a platform for information exchange. There are two main types of transactions: identity management transactions and model updating transactions. Identity management transactions are used for authentication. Model updating transactions are used to record models and enable sharing. Considering the latency and throughput bottlenecks facing these transactions, we suggest exploiting the InterPlanetary file system (IPFS) to implement off-chain storage if the size of the model is large. In this case, only the hash of the model and a uniform resource identifier (URI) are stored in the blockchain, where the URI represents a pointer to a file hosted in the IPFS.

D. P2P Network Layer

The P2P network layer consists of committee nodes and training nodes. It is based on Hyperledger Fabric, a consortium blockchain platform that supports smart contracts. As a distributed ledger, it stores the registration information and global model updates. In this article, one supernode is deployed for each domain, and these supernodes together form a committee. The supernodes, also called committee nodes, jointly maintain the blockchain ledger. The committee nodes verify transactions by

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means of consensus algorithms [e.g., practical Byzantine fault tolerance (PBFT) and Raft] and write them to the ledger after reaching consensus. The training nodes are divided into local training nodes and global training nodes. The local training nodes (e.g., drones) acquire local model updates based on local data and upload these updates to the model pool. The global training nodes are a subset of the committee nodes and are designed to honestly execute aggregation algorithms. Specifically, they compute global updates by using the local updates in the model pool in accordance with the aggregation rules written in the smart contracts.

E. BE-DHFL-Based Collaboration

In real-life scenarios such as automated factories, drones collaborate on complex tasks, such as assembly line control, shop floor monitoring, and cargo transportation. To improve efficiency, these tasks are split into multiple subtasks assigned to individual drones. We consider different workshops or factories as different domains. In the case of cargo transportation, for example, to achieve accurate cargo recognition, a drone needs to repeatedly engage in local learning to improve its recognition rate. However, the limited number of datasets collected for training by the drones in a single domain presents a bottleneck for their performance improvement. In addition, the training data may involve commercial secrets and cannot be shared with the public. In such a scenario, a cross-domain FL mechanism based on BE-DHFL can greatly improve drone performance by enabling collaborative learning across domains while safeguarding the privacy of the training data.

IV. Decentralized Horizontal Federated Learning Based on Blockchain

The proposed BE-DHFL system includes the following roles.

Blockchains: Private blockchains and a consortium blockchain are applied in this article. The private blockchains are used for the management of intradomain resources. The consortium blockchain is used for cross-domain authentication and global model updating.

Smart Contracts: The contracts used in this article include authentication contracts and updating contracts.

Domain Administrators: They act as the administrators of the domains and the maintainers of the consortium blockchain. In our scheme, the administrator of each domain is the corresponding KGC.

Drones: Drones need to be registered in both a private blockchain and the consortium blockchain. They are responsible for collecting data and participating in FL.

A. Federated Learning in BE-DHFL

In this article, we use decentralized smart contracts in place of a centralized aggregator. The blockchain-based aggregation algorithm (Algorithm 1) improves the security and auditability of the FL system. In addition, the constructed model is protected from poisoning attacks by an access mechanism that enables only authorized users to join the FL system. In the typical blockchain structure, each block contains header and body sectors. For simplicity, we divide the blocks into two types: authentication blocks and training blocks. Authentication blocks store cross-domain authentication information, and training blocks store local and global model updates. The length of a training block is $L = h + \delta_l \cdot N + \delta_g$, where h denotes the length of the block header, δ_l denotes the length of a local update, N denotes the number of local updates required to complete a round of learning, and δ_g denotes the length of the global model. Moreover, the space corresponding to the first part of the block body (with a size of $\delta_l \cdot N$) is called the *model pool*.

We stipulate that all participants actively participate in the local learning process and have access to the latest global model at any time. Each drone participating in FL trains an initial model using its local dataset and submits its local updates to its associated committee node for validation. When the committee nodes reach a consensus, the local updates are added to the model pool. To handle the frequent changes in the online/offline status of participants encountered in real-life scenarios, we set the number of required local updates for each learning round to N and the maximum wait time to $T_{\max}$. The updating contract is triggered when the number of local updates in the model pool satisfies num $\geq N$ or the current wait time satisfies time $\geq T_{\max}$. A global training node aggregates N updates in the model pool into a new global model in accordance with the aggregation rules in the smart contract. The resulting global model update is shared and verified among the committee nodes. After the committee nodes reach a consensus, the global model is stored in the current block and synchronized with the blockchain ledger. When a new round of training is conducted, the participants obtain the latest global update from their associated committee nodes. The abovementioned process is repeated until the global model reaches convergence.

B. Workflow of BE-DHFL

Our scheme is constructed on the basis of bilinear mapping and blockchain technology. The workflow of the proposed scheme is shown in Fig. 4. The global parameters $\{\mathbb{G}_1, g, p, H\}$ are set, where $\mathbb{G}_1$ is an elliptic curve multiplicative cyclic group of order p, g is the generator, and H is the hash function $H : \{0,1\}^* \rightarrow \mathbb{G}_1$. The KGC of each domain generates public–private key pairs of the form (pk_i, sk_i), where the private key $sk_i \in \mathbb{Z}_p$ is chosen randomly by KGC and the public key is $pk_i = g^{sk_i}$. The KGCs of all domains need to negotiate to deploy authentication and updating contracts before collaborative learning. A transaction will be initiated to record the domains involved when the process of contract creation is complete. Next, we will elaborate on the process of two drones from different domains participating in FL. The process of drone d_i^A from domain A participating in FL is described as follows.

Step 1: Local registration and global registration. First, d_i^A registers in the private blockchain of domain A, and then, d_i^A registers in the consortium blockchain with the assistance of KGC^A.

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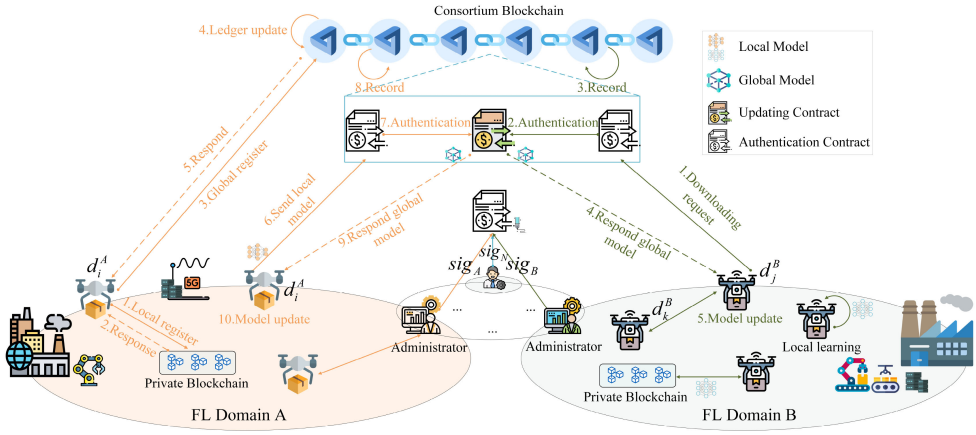


Fig. 4. Workflow in the proposed BE-DHFL framework.

Step 2: Local model update and upload. d_i^A uses its own data D_i for local learning and then broadcasts its local model m_i to the blockchain network through P2P.

Step 3: Authentication and global model update. The authentication contract is invoked by the committee nodes to authenticate d_i^A. If d_i^A is legitimate, then m_i will be added to the model pool for standby. Subsequently, once the number of models in the model pool reaches the threshold or the maximum wait time for a new round of training has been reached, the global training nodes call the updating contract to calculate the global model in accordance with the logic written in the contract. Finally, the newest global model $\mathcal{M}$ is returned to d_i^A.

Step 4: Local model update. d_i^A uses $\mathcal{M}$ to perform local training and obtain a local update m_{i+1}.

Under the assumption that drone d_j^B from domain B is a newly joined device that has completed registration, its participation in FL proceeds as follows.

Step 1: Download of the up-to-date global model. d_j^B initiates an authentication transaction with a global model download request to the consortium blockchain.

Step 2: Authentication. The committee nodes of the consortium blockchain invoke the authentication contract to verify the legitimacy of d_j^B; then, the updating contract is invoked and the model $\mathcal{M}$ is returned to d_j^B if the identity verification is passed.

Step 3: Local model update. d_j^B uses its local data D_j to train $\mathcal{M}$ and obtain a local update m_{j+1}.

C. Smart Contract Design

Two types of contracts are employed in this article, one defining the authentication management logic and the other defining the model aggregation logic. Notably, an authentication contract is a multisignature contract. We employ a broadcast multiple digital signature method with thresholds (t, n) [27]. The members involved are the signature request initiator, who is also the subsecret distributor SI, the participating domains P_i $(i = 1, 2, \ldots, n)$, the signature collector SC, and the signature verifier SV. Each P_i has its identifier DID_i. In the proposed scheme, SC and SV are the same entity, i.e., the contract deployer. The initiator of the contract creation request broadcasts the compiled bytecode σ of the contract to each P_i, where σ serves as the message to be signed. The multisignature construction process is as follows.

Initialization: All participants deliver their public keys pk_i as key shadows to SI. Next, SI randomly chooses a secret $k \in \mathbb{Z}_p$ and promulgates the secret shadow $K = g^k$. Next, the secret polynomial is constructed by choosing $t-1$ integers $a_1, \ldots, a_{t-1}$ to randomly generate a polynomial $f(x) = k + a_1 x + \cdots + a_{t-1} x^{t-1}$, and the subsecrets $d_i = f(\mathrm{DID}_i)$ are calculated. SI randomly selects an integer s as its master key and calculates $Q = g^s$. Then, an operation is performed with s and pk_i to obtain the subkeys $s_i = pk_i^s$, and $R_i = d_i \oplus H(s_l) \oplus H(s_m) \oplus \cdots \oplus H(s_j)$ are calculated, where l, m, and j represent the members involved in the secret sharing process. Finally, (DID_i, R_i) and Q are disclosed, where $i = 1, 2, \ldots, n$.

Subsecret recovery: Suppose that one participant wishes to recover the secret. First, this participant P_i sends its subkey s_i to the secret recoverer to estimate whether the equation $e(g, s_i) = e(Q, pk_i)$ holds. If it does not hold, the participant needs to be reverified. Conversely, when $t'(t \le t' \le n)$ participants are verified, the identity of each participant is confirmed, and each d_i is obtained by calculating $d_i = R_i \oplus H(s_l) \oplus H(s_m) \oplus \cdots \oplus H(s_j)$. Finally, d_i is sent to the corresponding P_i.

Single signature generation and verification: SI sends σ to each P_i; then, P_i computes $\mathrm{sig}_i = H(\sigma)^{d_i}$. Next, P_i sends its signature (σ, sig_i) on σ to SC and discloses the subsecret shadow $p_i = g^{d_i}$. After receiving (σ, sig_i), SC sets the number of valid signatures m to zero and verifies the validity of the signature by calculating $e(g, \mathrm{sig}_i) = e(H(\sigma), p_i)$.

Multisignature generation and verification: When $m \geqslant t$, SC calculates

$$\mathrm{MS} = \prod_{i=1}^{t} \mathrm{sig}_i^{\Delta_{i,P_x}(0)} = \prod_{i=1}^{t} H(\sigma)^{d_i \cdot \Delta_{i,P_x}(0)}$$
$$= H(\sigma)^{\sum_{i=1}^{t} d_i \cdot \Delta_{i,P_x}(0)} = H(\sigma)^{d}$$

where the $\Delta_{i,P}(x) = \prod_{j \in P, j \neq i} \frac{x-j}{i-j}$ are the Lagrange coefficients. Finally, SC sends (σ, MS) to SV as a multisignature. SV calculates whether the equation $e(g, \mathrm{MS}) = e(H(\sigma), K)$ holds;

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Algorithm 1: Blockchain-Based Model Aggregation Algorithm.

Input: drones d_i, global training nodes $n_{t,i}$, committee nodes $n_{c,i}$.
Output: optimized global model.

Initialization;
while New data D_i are generated **do**
  Each d_i performs a local model update;
  The $n_{c,i}$ authenticate d_i;
  Local updates are uploaded to the model pool;
  if The number of local updates in the model pool reaches the threshold **then**
    The $n_{t,i}$ update the global model;
  The $n_{c,i}$ generate a new block and propagate it;
  The d_i download the global update;
  end if
end while
Output the optimized global model $\mathcal{M}$.

if it holds, then the multisignature is valid. When the verification is passed, the updating contract will be deployed.

The authentication contract contains a registry with the registration logic shown in Algorithm 2 and the authentication logic shown in Algorithm 3. The updating contract is used to implement model aggregation, and its aggregation logic is shown in Algorithm 1.

D. Cross-Domain Authentication

In FL, the number of participants directly affects the performance of the final model. In practical scenarios, a domain (e.g., a factory) usually has a limited number of drones. The data collected by the drones are homogeneous in structure and type. To obtain a better model, the number of participants should be as high as possible. To this end, drones from different domains need to engage in collaborative learning. However, it is possible for malicious users to invade such a system or corrupt the model because a traditional FL system lacks an authentication mechanism. To address this shortcoming, we employ a consortium blockchain as a decentralized platform to provide registration and authentication services for cross-domain drones. Each KGC acts as the administrator for its own domain to perform cross-domain registration for drones. The registration algorithm is shown in Algorithm 2, where Addr_i is the account of d_i registered in the consortium blockchain, $\text{sig}_i = (\text{Addr}_i||h_i)^{sk_i}$ is a signature signed by KGC^i, $h_i = H(\text{ID}_i)$ is the hash value of the device identifier ID_i, T_i is a timestamp, and DID_i is the identifier of domain i. Suppose that d_i^A wishes to join the FL process; the steps taken by the system to authenticate this device are as follows.

1) d_i^A initiates a transaction $\text{Tran} = (\text{Addr}_i, \text{Addr}_c, \text{num}_i, ex)$. The payer is the account Addr_i, the payee is the authentication contract account Addr_c, the transaction amount is num_i, and the extra information is represented by $ex = (\text{join}||\text{ID}_i||\text{DID}_i)$, where join denotes the request to join the FL process.

Algorithm 2: Cross-domain Registration Algorithm.

Input: account address Addr_i, device identifier hash h_i, expiration date T_i, signature sig_i, signature domain identity DID_i.
Output: smart contract address.

Get public key $pk_i = pk[\text{DID}_i]$;
Let $m = \text{Addr}_i||h_i$;
Compute $e_1 = e(m, pk_i)$;
Compute $e_2 = e(g, \text{sig}_i)$; if $e_2 = e_1$ does not hold, output null;
Get domain object $D = \text{domain}[\text{DID}_i]$ and the index of the last registered device, $l = D.\text{endindex}$;
Set $D.\text{device}[l+1] = \text{Addr}_i$, $D.\text{hash}[l+1] = h_i$, $D.\text{time}[l+1] = T_i$, and $D.\text{status}[l+1] = \text{true}$;
Output the authentication contract address Wsc.

Algorithm 3: Cross-domain Authentication Algorithm.

Input: account address Addr_i, domain identifier DID_i.
Output: authentication result.

Get domain object $D = \text{domain}[\text{DID}_i]$;
Get index $i = D.\text{indexof}(\text{Addr}_i)$; if $i = \text{null}$ holds, output null;
Get status $= D.\text{status}[i]$, $T_i = D.\text{time}[i]$, and $h_i = D.\text{hash}[i]$;
Get $r = (\text{Addr}_i||\text{status}||T_i||h_i)$;
Output r.

2) The committee nodes verify the legitimacy of the transaction, verifying, for example, whether the signature of the transaction is legal and the balance of Addr_i is sufficient. If the transaction is legal, then the committee nodes call the authentication function in Algorithm 3. Finally, the verification result $r = (\text{Addr}_i||\text{status}||T_i||h_i)$ is returned. If Addr_i has been registered and the registration period is valid, then status $=$ true; otherwise, status $=$ false.
3) The committee nodes judge the verification result; if status $=$ false, then the transaction is discarded, whereas if status $=$ true, then the registration validity T_i will be judged. Let time denote the current timestamp. If $T_i <$ time, this indicates that the registration date is invalid, and the operation is terminated. If $T_i >$ time, then the registration date is valid, and the committee nodes calculate the hash of ID_i for d_i^A to obtain $h = H(\text{ID}_i)$. If $h = h_i$ holds, this means that d_i^A's identity is legal. Then, the model m_i uploaded by d_i^A will be added to the model pool.

E. Privacy-Enhanced FL

In FL, parameter updates in plaintext form are vulnerable to membership inference attacks. For this reason, we apply homomorphic encryption to prevent the global training nodes from obtaining global updates and prevent other participants from obtaining local updates by employing multiparty computation and adding random masks.

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Specifically, each participant i encrypts its local parameter g_i^t by using the homomorphic encryption key k to obtain $[[g_i^t]]_k$. Homomorphic encryption prevents the aggregator from acquiring g_i^t but does not prevent other participants from acquiring it because each participant has a decryption key. Therefore, we calculate $([[g_i^t]]_k + \varphi_i)$ to protect g_i^t. The mask φ_i is a preselected random number, and each mask factor φ_{ij} of the split is sent to each participant j through a secure channel after secret partitioning by using the (t, n) threshold method. When updating parameters, each participant calculates the sum $\varphi_i' = \sum_{j\in U} \varphi_{ji}$ of the mask factors of all participants (where U denotes all participants) and takes $([[g_i^t]]_k + \varphi_i)$ and φ_i' as the input for the updating contract. Let $S \subseteq U$ be a set of t participants who provide the sum of the mask factors. The encrypted global parameter is calculated as follows:

$$\begin{aligned}
&\sum_{i\in U}([[g_i^t]]_k + \varphi_i) - \sum_{i\in S}\varphi_i'\Delta_{i,S}(0)\\
&= \sum_{i\in U}([[g_i^t]]_k + \varphi_i) - \sum_{i\in S}(\sum_{j\in U}\varphi_{ji})\Delta_{i,S}(0)\\
&= \sum_{i\in U}([[g_i^t]]_k + \varphi_i) - \sum_{j\in U}(\sum_{i\in S}\varphi_{ji}\Delta_{i,S}(0))\\
&= \sum_{i\in U}([[g_i^t]]_k + \varphi_i) - \sum_{j\in U}(\varphi_j)\\
&= \sum_{i\in U}([[g_i^t]]_k = [[g^t]]_k)
\end{aligned}$$

where the $\Delta_{i,S}(x) = \prod_{j\in S, j\neq i}\frac{x-j}{i-j}$ are the Lagrange coefficients. Homomorphic encryption makes it infeasible for an aggregator (e.g., a global training node) to obtain any updates in plaintext form. When the numbers of both malicious collaborators and training rounds are fewer than t, the local updates are unavailable to other participants, thereby preventing attacks, such as membership inference and GAN attacks, against the local updates.

V. Security Analysis and Performance Evaluation

A. Security Analysis

In this article, the applied consortium blockchain establishes trusted communication among cross-domain drones. To protect model security, we utilize smart contracts to aggregate local updates and homomorphic encryption to encrypt the model parameters.

1) Secure Data Management: Throughout the learning process, the participants share data models rather than raw data; therefore, they have absolute control over their data. The model parameters are protected by homomorphic encryption to prevent leakage or membership inference attacks on the model during transmission. In addition, the consortium blockchain ensures data security and reliability through a series of encryption and signature algorithms.

2) Cross-Domain Authentication: To prevent malicious users from destroying the constructed model by means of poisoning attacks, we apply a multisignature smart contract to realize authentication and dynamic management of cross-domain drones. The multisignature mechanism is based on a threshold method to achieve controlled security. By introducing an access mechanism into FL and using blockchain technology for dynamic identity management, we eliminate intrusion and improve the security of the system.

3) Decentralization: A decentralized smart contract is applied instead of a centralized server for model aggregation. Traditional centralized servers present the risks of a single point of failure and data leakage, while smart contracts operate on the basis of a blockchain that is jointly maintained by multiple nodes. The encryption algorithms used in blockchains can protect data security and resist common distributed denial of service (DDoS) attacks.

To provide a broader context, we compare the proposed scheme with existing FL schemes under various security metrics. The results are shown in Table I.

B. Experimental Setting

In our experiments, we simulate a scenario in which FL is performed across two domains and all devices are connected via a local area network. We use two virtual machines acting as KGCs, with an Intel Core i7-7700 CPU @3.6 GHz, 4 GB of RAM, and the Ubuntu 18.04 64-bit OS. The Java Pairing-Based Cryptography Library and Hyperledger Fabric are applied to conduct the experiments. Moreover, the Java Development Kit (JDK) is installed, and consensus is reached in Hyperledger Fabric by means of the PBFT algorithm. Our FL experiments are based on digital images from the federated EMNIST dataset, which includes 341 873 training samples and 40 832 test samples. The dataset is divided into ten classes representing numbers from 0 to 9, and the size of the images is 28 $\times$ 28. In each training epoch, samples are randomly selected from the dataset. We leverage a convolutional neural network (CNN) model that is maintained in a distributed manner through blockchain technology. The designed network model possesses hidden layers for feature extraction and fully connected layers for classification. More specifically, the CNN model contains two convolutional layers with 32 and 64 channels. After each convolutional layer, a 2 $\times$ 2 max pooling layer is applied to reduce the data dimensionality and improve the learning efficiency. In addition, two fully connected layers are connected to the channels; the first uses the rectified linear unit activation function and has 256 units, while the second uses the softmax activation function and has ten units. The hyperparameters of the CNN model are set as follows. The default learning rate is 0.15, and the default batch size is 64. The default number of local epochs is 30, and the default number of global epochs is three. The default number of participants is ten. Moreover, the default number of domains is four. To evaluate the feasibility of the proposed scheme, we use a Raspberry Pi 4 Model B to simulate a drone. The configuration of the Raspberry Pi is as follows: Broadcom BCM2711, quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5 GHz, 8 GB of RAM, and 64 GB of ROM.

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TABLE I
COMPARISONS OF DIFFERENT SCHEMES UNDER VARIOUS SECURITY METRICS

Metric	[8]	[15]	[20]	[22]	[23]	BE-DHFL
Cross-domain authentication	No	No	No	No	No	Yes
Privacy preservation	Yes	Yes	Yes	Yes	Yes	Yes
Decentralization	No	No	Yes	Yes	Yes	Yes
Traceability	Yes	Yes	Yes	Yes	Yes	Yes
DDoS attacks	Yes	Yes	Yes	Yes	Yes	Yes
Poisoning attacks	Yes	No	No	Yes	Yes	Yes
Membership inference attacks	No	Yes	Yes	No	No	Yes
Reconstruction attacks	No	No	No	No	No	Yes

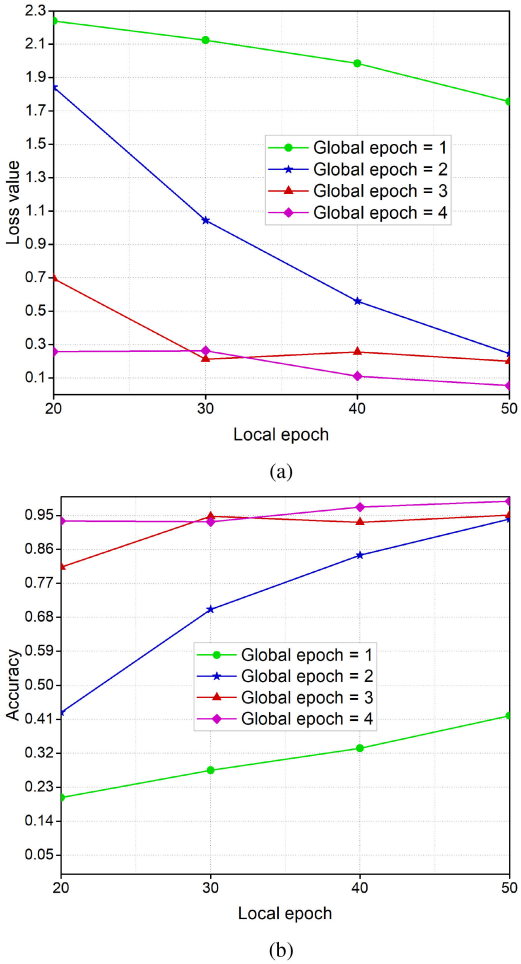


Fig. 5. Impact of the number of local epochs on the learning performance under various numbers of global epochs.

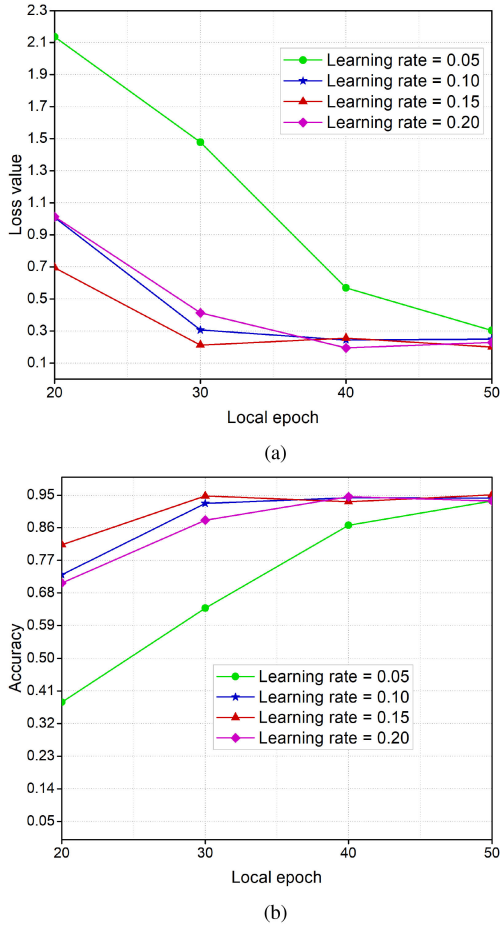


Fig. 6. Impact of the number of local epochs on the learning performance under various learning rates.

C. Performance of BE-DHFL

Because system initialization and registration are performed in advance, the drones need to undertake only a small amount of computation during the cross-domain authentication phase. The time complexity and space complexity of both Algorithms 2 and 3 are $O(1)$. In Algorithm 2, two bilinear pairings and one blockchain write operation need to be executed, and the total time consumed is approximately 0.28 s. In Algorithm 3, one blockchain read operation needs to be executed, and the time consumed is approximately 0.02 s. Therefore, the proposed algorithms are friendly to resource-constrained drones. When the network communication delay is neglected, it takes approximately 187.79 s to complete three rounds of learning, after which the model accuracy can reach 0.94. Consensus needs to be reached among the committee nodes before data can be written to the blockchain. It is possible to improve the efficiency of transaction processing by deploying the committee nodes on servers with higher computing power. In addition, by optimizing the consensus algorithm, it is possible to improve the efficiency of writing data and thus guarantee the efficiency of FL.

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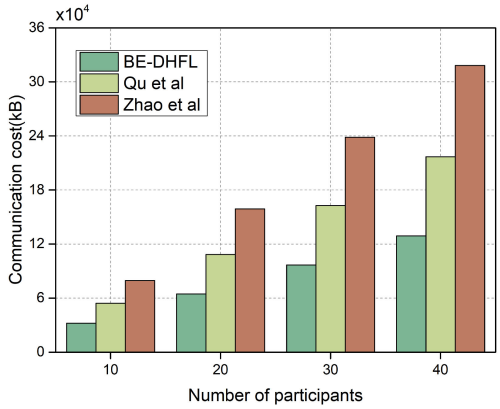


Fig. 7. Communication cost of FL with different numbers of participants.

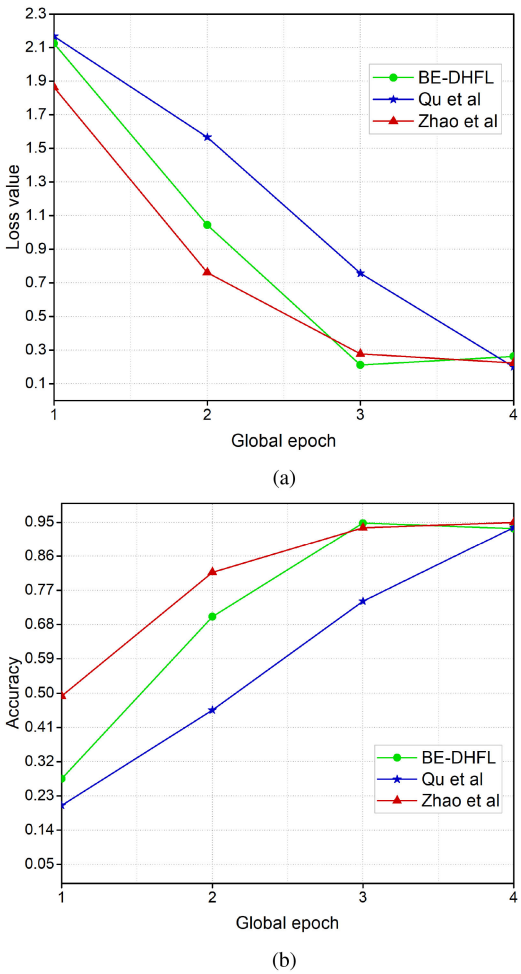


Fig. 8. Impact of the number of global epochs on the learning performance.

For cross-domain authentication, we calculate only the size of messages related to authentication. First, d_i^A initiates a 376-B transaction. After receiving the transaction, a committee node submits a 232-B identity query request to the consortium blockchain, and a result with a size of 792 B is returned. The committee node verifies the result to complete cross-domain authentication. The total overhead is 1400 B. In our scheme, the average model size is 913.3 kB. The transmission time of the model depends on the network bandwidth. Suppose that the drones are connected through a 5G or Wi-Fi network with a bandwidth of 5 MB/s. In this case, the transmission time needed for one model is 0.1783 s, which is friendly to resource-constrained drones.

In our experiment, *local epoch* = 1 means that each participant locally trains the model once using all training samples, and *global epoch* = 1 means that one round of model aggregation is performed. As shown in Fig. 5(a), as the number of local epochs increases when the number of global epochs is fixed, the loss value shows a decreasing trend; however, if the number of global epochs is small, the loss remains at a high level. As the number of global epochs increases when the number of local epochs is fixed, the loss value shows a larger decrease. Fig. 5(b) shows how the model accuracy is affected by the number of local epochs under different numbers of global epochs. As shown in the figure, even if the participants perform multiple epochs locally, the accuracy is not high if the number of global epochs is small. As the numbers of both local and global epochs increase, the accuracy appears to rise sharply. When the number of local epochs reaches 50 and the number of global epochs reaches four, an accuracy of 0.98 is achieved. However, the number of global epochs has a greater impact on accuracy. If trained only locally, the model will show poor performance due to the limited amount of local data available and the lack of data diversity. A better model will be available to each participant if more trainers join the FL process to train a global model. When we run the same FL training process on a Raspberry Pi 4 using 5000 randomly selected digital images from EMNIST, it takes 128 s to train the model in each local epoch. For 30 local epochs, the time spent is 64 min. Therefore, it is feasible for idle drones equipped with backup power to join the FL process.

Fig. 6 shows the learning performance under different numbers of local epochs and different learning rates. The learning rate controls the updating of the model weights, which affects the efficiency and speed of model convergence. Fig. 6(a) shows how the loss value changes when the number of local epochs is fixed and the learning rate varies. When the learning rate is fixed and the number of local epochs varies, there exists an optimal number of local epochs such that the loss value is minimized. For example, the loss value is the smallest when the learning rate is 0.20 and the number of local epochs is 40. It can be concluded that both the learning rate and the number of local epochs affect the model performance, and a similar trend is observed in Fig. 6(b). Therefore, it is necessary to find a balance between the learning rate and the number of local epochs to obtain the optimal model with the minimum cost.

D. Performance Comparisons

We compare the proposed scheme with the FL schemes of Qu *et al.* [8] and Zhao *et al.* [20] in the experimental setting described in Section V-B. Fig. 7 gives the communication overheads of a round of FL performed with different numbers of participants. When the number of participants is ten, the

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communication overheads of BE-DHFL, Qu *et al.* [8] and Zhao *et al.* [20] are 32 266.6 kB, 54 238.2 kB, and 79 514.2 kB, respectively. As the number of participants increases, BE-DHFL shows a communication advantage, indicating that BE-DHFL offers better communication performance. Fig. 8 presents the effects of the number of global epochs on the performance of the different schemes. It can be seen from this figure that under all schemes, the model performance improves substantially as the number of global epochs increases. When the number of global epochs reaches four, the performance gap among the schemes starts to narrow. Throughout the training process, the BE-DHFL scheme outperforms the scheme of Zhao *et al.* [20] Due to the normalization technique introduced by Qu *et al.* [8] their scheme achieves better model performance than BE-DHFL during the training process; however, because the participants are not authenticated, an attack by illegal participants may destroy the usability of the model. Therefore, the proposed BE-DHFL scheme maintains good model performance while improving the robustness of the FL system.

VI. Conclusion

This article demonstrates a decentralized cross-domain FL framework named BE-DHFL for 5G-enabled drones based on blockchain technology. We employ threshold multisignature contracts to provide dynamic authentication services for cross-domain drones to eliminate barriers to collaborative learning. Decentralized smart contracts are applied to achieve model aggregation, thereby solving the security problems faced by centralized servers, such as the risks of a single point of failure and data leakage. In addition, we exploit homomorphic encryption and multiparty computation to resist attacks against local updates. The feasibility of applying the proposed scheme to drones is demonstrated by security and experimental analyses. Nevertheless, the proposed scheme also has limitations. For example, the model is not compressed, which places pressure on the communication capabilities of resource-constrained devices. Moreover, no incentive mechanism is implemented, meaning that participants cannot be effectively encouraged to join in the FL process. In the future, we will explore effective solutions for these two shortcomings.

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